ERGOTAMINE + CAFFEINE

Ergotism risk

Contraindicated in pregnancy

CHEMICAL NAME Ergotamine tartrate: (5′α)-12′-hydroxy-2′-methyl-5′-(phenylmethyl)ergotaman-3′,6′,18-trione; caffeine: 1,3,7-trimethylpurine-2,6-dione

CLASS Antimigraine combination: ergot alkaloid (vasoconstrictor) + methylxanthine

TRADE NAMES Cafergot · Migril · Ergomar (ergotamine)

THERAPEUTIC USES / INDICATIONS

- Acute migraine attacks not responding to simple analgesics or triptans.
- Cluster headache (short-term).

ADMINISTRATION

- **PO** tablets at first sign of attack.
- **PR** suppositories (better absorption, useful with vomiting).
- Strict weekly maximum doses.

DOSAGES

- **Tablets (1 mg ergotamine/100 mg caffeine):** 2 tablets at onset, then 1 every 30 min if needed.
- Max 6 mg ergotamine per attack and 10 mg per week.
- Do not repeat within 4 days; max 2 attacks treated per month (practical limit).

PHARMACOKINETICS

- **Onset:** 30 min–1 h (PR faster).
- Poor oral absorption, improved by caffeine; extensive first-pass metabolism (CYP3A4).
- Excreted mainly in bile. **t½:** 2 h, but vasoconstriction lasts 24 h.

PHARMACODYNAMICS

- Ergotamine: partial agonist at **5-HT1B/1D**, α-adrenergic and dopamine receptors → constriction of dilated cranial vessels.
- Caffeine: enhances absorption and adds vasoconstriction.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, vomiting (common), abdominal pain.
- Muscle cramps, numbness, tingling in fingers and toes.
- Restlessness, insomnia (caffeine).

ADVERSE EFFECTS (SERIOUS)

- Ergotism: severe peripheral vasoconstriction → cold, painful, pale limbs, gangrene.
- Myocardial ischaemia; arrhythmias.
- Medication-overuse headache; fibrosis with chronic use.
- Uterine contractions.

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (oxytocic) and breastfeeding.
- Coronary, peripheral vascular or cerebrovascular disease; uncontrolled hypertension.
- Severe hepatic or renal impairment; sepsis.
- Use with strong CYP3A4 inhibitors or within 24 h of a triptan.

KEY INTERACTIONS

- **Clarithromycin, erythromycin, azole antifungals, HIV protease inhibitors:** acute ergotism — contraindicated.
- **Triptans, β-blockers:** ↑ vasoconstriction.

NURSING MANAGEMENT

- Check pregnancy status and cardiovascular history.
- Record doses carefully — enforce weekly maximum.
- Check limbs for coldness, numbness, colour change, pulses.
- Assess headache response; give antiemetic if needed.
- Review interacting antibiotics and antifungals.

PATIENT EDUCATION

- Never exceed the dose per attack or per week.
- Report cold, numb or painful fingers or toes immediately.
- Do not use if pregnant or planning pregnancy.
- Avoid triptans within 24 h; reduce coffee intake.

KEY POINTS

- Ergot vasoconstrictor + caffeine for migraine.
- Ergotism risk; check limbs.
- Max 6 mg/attack, 10 mg/week.
- Never with macrolides or azoles; not in pregnancy.